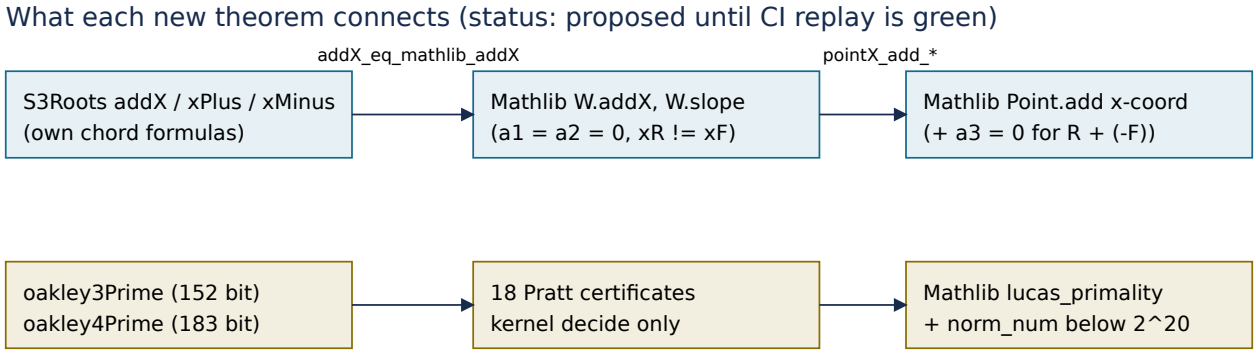
Lean bridge for S3Roots and Pratt certificates for the Oakley orders

2026-10-29. Origin: CFSG exposure audit (crypto-autoresearcher PR #2690). Status: **proposed**; the Lean builds are verified only by the CI runs named in the PR, not locally (no Lean toolchain could be installed in this session).



Task 1: S3Roots and Mathlib's group law

Mathlib pin: d13f23b7…/v4.34.1 (unchanged). Lemmas are in papers/rootfinding-decomposition/lean/S3Roots.lean.

Theorem	Establishes	Hypothesis-delta (binders the paper's sentence does not state)	Does not establish
addX_eq_mathlib_addX	addX xR yR xF yF = W.addX xR xF (W.slope xR xF yR yF)	DecidableEq K; $W.a_1=0$, $W.a_2=0$ (paper: short Weierstrass). No curve equation, no nonsingularity.	Anything about a_4,a_6; the equal-x branch.
pointX_add_eq_xPlus	x-coordinate of Mathlib some R + some F is xPlus	As above, plus W.Nonsingular at both points (Mathlib Point requires it); $xR \neq xF$.	$R = \pm F$ cases.
pointX_add_neg_eq_xMinus	x-coordinate of some R + - some F is xMinus	As above plus $W.a_3=0$ (negY = -y).	Same.
S3_factor_mathlib	S3_factor for Mathlib points, with equations derived from Nonsingular; conjunction with the two x-coordinate facts	$a_1=a_2=a_3=0$; S3 $W.a_4$ $W.a_6$.	S_m, $m \geq 4$, remains unformalized.

Axiom audit: the four theorems are added to AxiomAudit.lean and the CI count in rootfinding-s3roots-lean.yml is raised from 3 to 7. The expected result is only propext, Classical.choice, Quot.sound; this is unverified until CI is green.

Task 2: primality of the Oakley orders

research/weak_curve_leaderboard_20261012/lean/OakleyPrimality.lean.

Theorem	Establishes	Hypothesis-delta	Does not establish
Pratt.pratt	p<2^256, factor list L of primes with product p-1, Fermat and witness checks by powMod imply Nat.Prime p	p < 2^256 (bounds the exponent for powMod), not in textbook Pratt.	Anything beyond primality.
Pratt.powModAux_cast	powModAux computes acc·b^e in ZMod m for e < 2^fuel	none	
oakley3Prime_prime, oakley4Prime_prime	Nat.Prime of the two integers via 18 certificates (factors below 2^20 by norm_num)	None beyond the numerals in OakleyOrders.lean.	That they are RFC point orders; that the curves are as described.

The same data is in validation/oakley_pratt_certificates.json; the Rust checker src/bin/check_pratt_certificates.rs (num-bigint) passed locally: 18 certificates, 49 primes. Certificates were produced with sympy factorisation of p-1 (an untrusted producer: only the checks matter). decide +kernel uses kernel evaluation, not native_decide.

Open

Lean builds, leanchecker --fresh and axiom output pending CI; if decide +kernel is too slow or a tactic fails, the certificate JSON and Python checker remain the fallback evidence.